Experiment No.10

Aim: To perform web scraping using the Scrapy tool.

THEORY: -

Web scraping is the process of automatically extracting useful information from websites. Normally, a user opens a website and reads information manually. In web scraping, a program automatically:

For example, an online product page may contain:

- Product Name
- Price
- Rating
- Availability
- Product URL

A web scraping program can collect this information and store it in CSV, JSON, or database format.

Tool Used – Scrapy

Scrapy is an open-source Python framework used for web crawling and web scraping.

It provides components for:

- Sending requests to web pages
- Downloading web pages
- Extracting required information
- Following links
- Processing scraped data
- Storing data in structured formats

Note: Scrapy is different from a normal web browser. A browser displays the page for a human, while Scrapy automatically requests and processes pages.

Web Scraping Using Scrapy – Collecting Product Information from a Website

Suppose an online bookstore contains information about several books. A company wants to analyze book prices and ratings.

Manually collecting this information from hundreds of pages would take considerable time. Therefore, a Scrapy spider can be created to automatically collect:

Field	Description
Book Name	Name of the book
Price	Price of the book
Rating	Customer rating
Availability	Stock status

The collected data can then be used for further data analysis and visualization.

Web Scraping Process

WEBSITE
↓
Send HTTP Request
↓
Download Web Page
↓
Scrapy Spider
↓
Locate HTML Elements
↓
Extract Required Data
↓
Clean / Process Data
↓
Store in CSV / JSON / DB
↓
Data Analysis & Mining

Example

Suppose the webpage contains:
Book: Data Science Basics
Price: ₹450
Rating: 4.5
Availability: In Stock

Scrapy extracts these values and produces:

Book Name	Price	Rating	Availability
Data Science Basics	450	4.5	In Stock

Important Components

Spider
Defines what website to crawl and what information to extract.

Item
Defines the fields that will be collected.

Selector
Used to locate elements in HTML.

Pipeline
Processes and stores scraped data.

Data Analysis

Once the data has been collected, students can perform simple analysis.

Example Questions

1. Which book has the highest price?
2. Which book has the highest rating?
3. What is the average book price?
4. How many books are in stock?
5. Is there any relationship between price and rating?

For example:

Scraped Data
↓
Data Cleaning
↓
Data Analysis
↓
Visualization
↓
Useful Information

Possible Visualization

The scraped data can be represented using:

Bar Chart

Book Name vs Price

Pie Chart

Available vs Out-of-Stock Books

Scatter Plot

Price vs Rating

Advantages of Scrapy

- Automates data collection.
- Can crawl multiple web pages.
- Saves time compared with manual collection.
- Supports CSV and JSON output.
- Suitable for large-scale web crawling.
- Can follow links between pages.
- Provides tools for data processing.

Limitations

- Websites may change their HTML structure.
- Some websites use JavaScript to dynamically load data.
- Websites may restrict automated requests.
- Scraping large numbers of pages can put load on a server.
- Data may contain duplicates or errors.
- Legal and website-specific restrictions must be considered.

Ethical Considerations:

Before scraping:

- Check the website's terms of service.
- Respect rate limits.
- Avoid collecting personal or sensitive information.
- Do not bypass authentication, CAPTCHAs, or access controls.
- Prefer websites specifically designed for scraping practice.

Conclusion

Thus, web scraping using Scrapy was studied and demonstrated. The process automatically extracts useful information from web pages, converts it into structured data, and makes the data available for further analysis and visualization.